Week 2 progressive hints

Use one hint at a time. Return to the question after each hint before reading the next one.

W02-Q01-PA

1. Check the inner dimensions before multiplying: $\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×2.
2. Form each entry by taking one row of $\mathbf{A}$ times one column of $\mathbf{B}$.
3. For the associativity check, first find $\mathbf{Bu}$, then compare the resulting vector with the product of $\mathbf{AB}$ and $\mathbf{u}$.

W02-Q01-PB

1. The reversed product also exists, but its outer dimensions are different.
2. Each row of $\mathbf{BA}$ is a linear combination of the two rows of $\mathbf{A}$, with coefficients from the corresponding row of $\mathbf{B}$.
3. Transpose your 3×3 product. Separately multiply the 3×2 matrix $\mathbf{A}^\mathsf{T}$ by the 2×3 matrix $\mathbf{B}^\mathsf{T}$, then compare both the dimensions and entries.

W02-Q02-PA

1. For a 2×2 matrix, compute $ad - bc$ first.
2. Because the determinant is non-zero, interchange the diagonal entries, change the signs of the off-diagonal entries, and divide by the determinant.
3. Multiply $\mathbf{C}$ by the proposed inverse; the diagonal entries of the product should be one and the off-diagonal entries zero.

W02-Q02-PB

1. Left-multiply both sides by $\mathbf{C}^{-1}$, so $\mathbf{X} = \mathbf{C}^{-1}\mathbf{D}$ for the displayed right-hand-side matrix $\mathbf{D}$.
2. Treat the two columns of $\mathbf{D}$ as two right-hand sides and keep the two solution columns in the same order.
3. Verify the full matrix product and connect uniqueness to the non-zero determinant.

W02-Q03-PA

1. Write the 4×5 augmented matrix and use the first row to eliminate x from rows 2, 3 and 4.
2. One convenient start is $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$, followed by $R_4 \leftarrow R_4 - 3R_1$.
3. Swap the new rows 2 and 3 to obtain a pivot of one in the y column; then eliminate y, eliminate z, and back-substitute from w.

W02-Q04-PA

1. The coefficient row of the third equation is the sum of the first two coefficient rows.
2. Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, and then subtract the new row 2 from row 3.
3. The final augmented entry depends on k. Decide when it is zero.

W02-Q04-PB

1. Compare $\operatorname{rank}(\mathbf{A})$ with $\operatorname{rank}([\mathbf{A} \mid \mathbf{b}])$ before counting free variables.
2. In the consistent case there are three unknowns and only two pivots. Let $z = t$.
3. Use the second echelon equation to find y, then the first to find x. Substitute the resulting expressions into each original equation to perform the requested verification.

W02-Q05-PA

1. To make $\mathbf{PQ} = \mathbf{0}_{2\times 2}$, arrange for every column of $\mathbf{Q}$ to be sent to zero by $\mathbf{P}$.

2. Let $\mathbf{P}$ keep the first coordinate and kill the second. Choose a rank-one $\mathbf{Q}$ whose non-zero output lies in the second coordinate, but which does not kill the first coordinate.
3. Compute the reverse product separately; do not infer it from the forward product.

W02-Q06-PA

1. Use the multiplication rule separately on the branches E and E^c, beginning with $\mathbb{P}(E^c)$.
2. The conditional probabilities of M^c are the complements of the supplied conditional probabilities of M.
3. Arrange the four joint probabilities in a table and obtain each marginal by summing its row or column.

W02-Q06-PB

1. Use the M column total as the denominator of $\mathbb{P}(E \mid M)$ and the M^c column total as the denominator of $\mathbb{P}(E \mid M^c)$.
2. Independence would require the conditional probability of E to equal its marginal probability in either conditioning group.
3. Describe the direction of association by comparing the two conditional proportions, without claiming that the context establishes causation.

W02-Q07-PA

1. The three session events form a partition, so there are three mutually exclusive routes to C.
2. Multiply $\mathbb{P}(C \mid R_i)$ and $\mathbb{P}(C^c \mid R_i)$ by $\mathbb{P}(R_i)$ to fill each row.
3. Add the C column to obtain $\mathbb{P}(C)$, and check all six cells sum to one.

W02-Q07-PB

1. The requested conditioning direction is the reverse of the probabilities supplied in the question.
2. Divide each entry in the C column by the C column total.
3. A valid posterior distribution over the three sessions must sum to one; then compare the three values.

W02-Q08-PA

1. Recover $\mathbb{P}(A \cap B)$ from the addition rule for $\mathbb{P}(A \cup B)$.
2. Divide the intersection by $\mathbb{P}(B)$ to find $\mathbb{P}(A \mid B)$.
3. Compare the intersection with $\mathbb{P}(A)\mathbb{P}(B)$ and the conditional probability with $\mathbb{P}(A)$. For disjointness, compare the intersection with zero.

W02-Q09-PA

1. Treat D and D^c as a partition and find $\mathbb{P}(S)$.
2. For the negative signal, complement both conditional signal probabilities and find $\mathbb{P}(S^c)$ before applying Bayes' rule again.
3. To interpret the result, distinguish $\mathbb{P}(S \mid D)$ from $\mathbb{P}(D \mid S)$, and account for the large D^c group.

W02-Q10-PA

1. Both logarithm arguments must be positive.
2. Combine the logarithms into the logarithm of a product, then use the fact that log is one-to-one on its domain.
3. Solve the resulting quadratic and test every root against the original domain.

W02-Q10-PB

1. Evaluate $f(1)$, $f'(1)$ and $f''(1)$.
2. Substitute those values into $f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2$.
3. The displacements are -0.1 and 0.1. Squaring removes the sign, while $f''(1) < 0$ fixes the sign of the quadratic correction.

W02-Q10-PC

1. "At least one" corresponds to a union. Apply De Morgan's law to write the complement as the event that neither named student is selected.
2. There are $\binom{5}{2}$ unordered pairs in total. For the complement, choose both representatives from the other three students.
3. For a direct check, partition the favourable pairs into the disjoint cases "exactly one named student" and "both named students".